

Universidad Nacional Autónoma de México



Facultad de Ingeniería

División de Ingeniería Eléctrica



Laboratorio Computación Gráfica e Interacción
Humano-Computadora (6590)

Práctica 3: Modelado geométrico

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Calificación: _____

1.Ejercicios

Para esta práctica se buscó crear una pirámide conocida como pyraminx. Para esto se tuvieron que hacer bastantes modificaciones al código para hacerlo funcionar como debería. Para comenzar, se decidió “ahorrar” recursos, por lo que se comenzó por eliminar la llamada a crear otras figuras y solo se crearon las pirámides.

```
291
292      CrearPiramideTriangular();
293      CreateShaders();
294
```

Posterior a eso se buscó crear una pirámide base, en este caso, debido a que se trabajó en un entorno color negro, se hizo una pirámide base color blanca.

```
332      // Triángulo base color blanco
333      model = glm::mat4(1.0f);
334      color = glm::vec3(1.0f, 1.0f, 1.0f);
335      model = glm::scale(model, glm::vec3(1.5f, 1.5f, 1.5f));
336      model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
337      model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
338      model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
339      model = glm::translate(model, glm::vec3(0.0f, 0.34f, 0.287f));
340      glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
341      glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
342      glUniform3fv(uniformColor, 1, glm::value_ptr(color));
343      meshList[0]->RenderMesh();
```

Y una vez que se tuvo la base, se añadieron las caras con algunos de mis colores favoritos, comenzado por el color morado.

```
345      // Cara morada
346      color = glm::vec3(0.424f, 0.275f, 0.459f);
347
348      model = glm::mat4(1.0);
349      model = glm::translate(model, glm::vec3(0.0f, -0.01f, -0.05f));
350      model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
351      model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
352      model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
353      model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
354      glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
355      glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
356      glUniform3fv(uniformColor, 1, glm::value_ptr(color));
357      meshList[0]->RenderMesh();
358
359      model = glm::mat4(1.0);
360      model = glm::translate(model, glm::vec3(0.5f, -0.01f, -0.05f));
361      model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
362      model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
363      model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
364      model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
365      glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
366      glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
367      glUniform3fv(uniformColor, 1, glm::value_ptr(color));
368      meshList[0]->RenderMesh();
```

```

370     model = glm::mat4(1.0);
371     model = glm::translate(model, glm::vec3(-0.5f, -0.01f, -0.05f));
372     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
373     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
374     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
375     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
376     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
377     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
378     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
379     meshList[0]->RenderMesh();
380
381     model = glm::mat4(1.0);
382     model = glm::translate(model, glm::vec3(-0.25f, 0.4f, 0.09f));
383     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
384     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
385     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
386     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
387     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
388     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
389     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
390     meshList[0]->RenderMesh();

```

```

392     model = glm::mat4(1.0);
393     model = glm::translate(model, glm::vec3(0.25f, 0.4f, 0.09f));
394     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
395     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
396     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
397     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
398     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
399     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
400     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
401     meshList[0]->RenderMesh();
402
403     model = glm::mat4(1.0);
404     model = glm::translate(model, glm::vec3(0.0f, 0.81f, 0.235f));
405     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
406     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
407     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
408     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
409     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
410     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
411     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
412     meshList[0]->RenderMesh();

```

```

414     model = glm::mat4(1.0);
415     model = glm::translate(model, glm::vec3(0.25f, -0.13f, -0.1f));
416     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
417     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
418     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
419     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
420     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
421     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
422     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
423     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
424     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
425     meshList[0]->RenderMesh();
426
427     model = glm::mat4(1.0);
428     model = glm::translate(model, glm::vec3(-0.25f, -0.13f, -0.1f));
429     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
430     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
431     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
432     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
433     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
434     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
435     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
436     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
437     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
438     meshList[0]->RenderMesh();

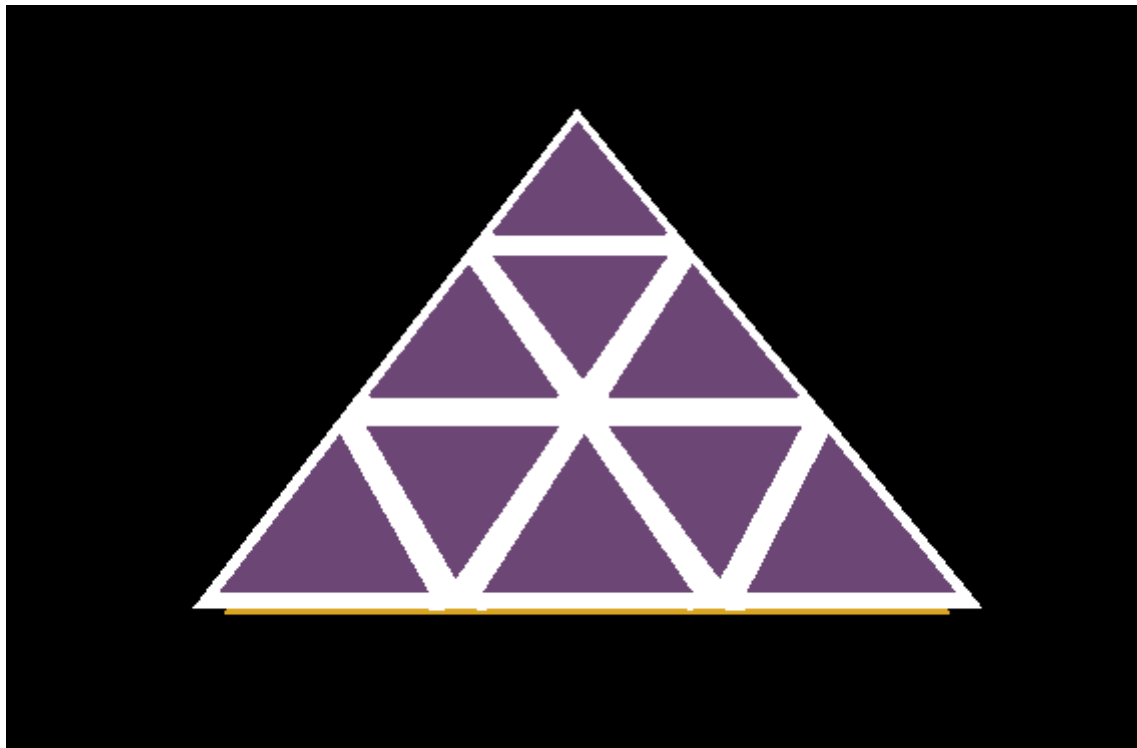
```

```

440     model = glm::mat4(1.0);
441     model = glm::translate(model, glm::vec3(0.0f, 0.29f, 0.05f));
442     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
443     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
444     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
445     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
446     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
447     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
448     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
449     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
450     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
451     meshList[0]->RenderMesh();

```

Al ejecutor el código la cara morada se ve de la siguiente manera.



Posterior a esa se añadió la cara verde.

```

453     // Cara verde
454     color = glm::vec3(0.133f, 0.545f, 0.133f);
455
456     model = glm::mat4(1.0);
457     model = glm::translate(model, glm::vec3(0.28f, -0.01f, 0.43f));
458     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
459     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
460     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
461     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
462     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
463     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
464     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
465     meshList[0]->RenderMesh();
466
467     model = glm::mat4(1.0);
468     model = glm::translate(model, glm::vec3(0.53f, -0.01f, 0.0f));
469     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
470     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
471     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
472     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
473     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
474     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
475     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
476     meshList[0]->RenderMesh();

```

```

478     model = glm::mat4(1.0);
479     model = glm::translate(model, glm::vec3(0.03f, -0.01f, 0.86f));
480     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
481     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
482     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
483     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
484     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
485     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
486     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
487     meshList[0]->RenderMesh();
488
489     model = glm::mat4(1.0);
490     model = glm::translate(model, glm::vec3(0.28f, 0.4f, 0.135f));
491     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
492     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
493     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
494     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
495     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
496     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
497     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
498     meshList[0]->RenderMesh();
499
500     model = glm::mat4(1.0);
501     model = glm::translate(model, glm::vec3(0.03f, 0.4f, 0.575f));
502     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
503     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
504     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
505     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
506     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
507     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
508     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
509     meshList[0]->RenderMesh();
510

```

```

511     model = glm::mat4(1.0);
512     model = glm::translate(model, glm::vec3(0.03f, 0.81f, 0.28f));
513     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
514     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
515     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
516     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
517     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
518     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
519     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
520     meshList[0]->RenderMesh();
521
522     model = glm::mat4(1.0);
523     model = glm::rotate(model, -120 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
524     model = glm::translate(model, glm::vec3(0.175f, 0.29f, -0.28f));
525     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
526     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
527     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
528     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
529     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
530     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
531     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
532     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
533     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
534     meshList[0]->RenderMesh();
535
536     model = glm::mat4(1.0);
537     model = glm::rotate(model, -120 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
538     model = glm::translate(model, glm::vec3(-0.07f, -0.13f, -0.42f));
539     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
540     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
541     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
542     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
543     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
544     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
545     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
546     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
547     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
548     meshList[0]->RenderMesh();
549

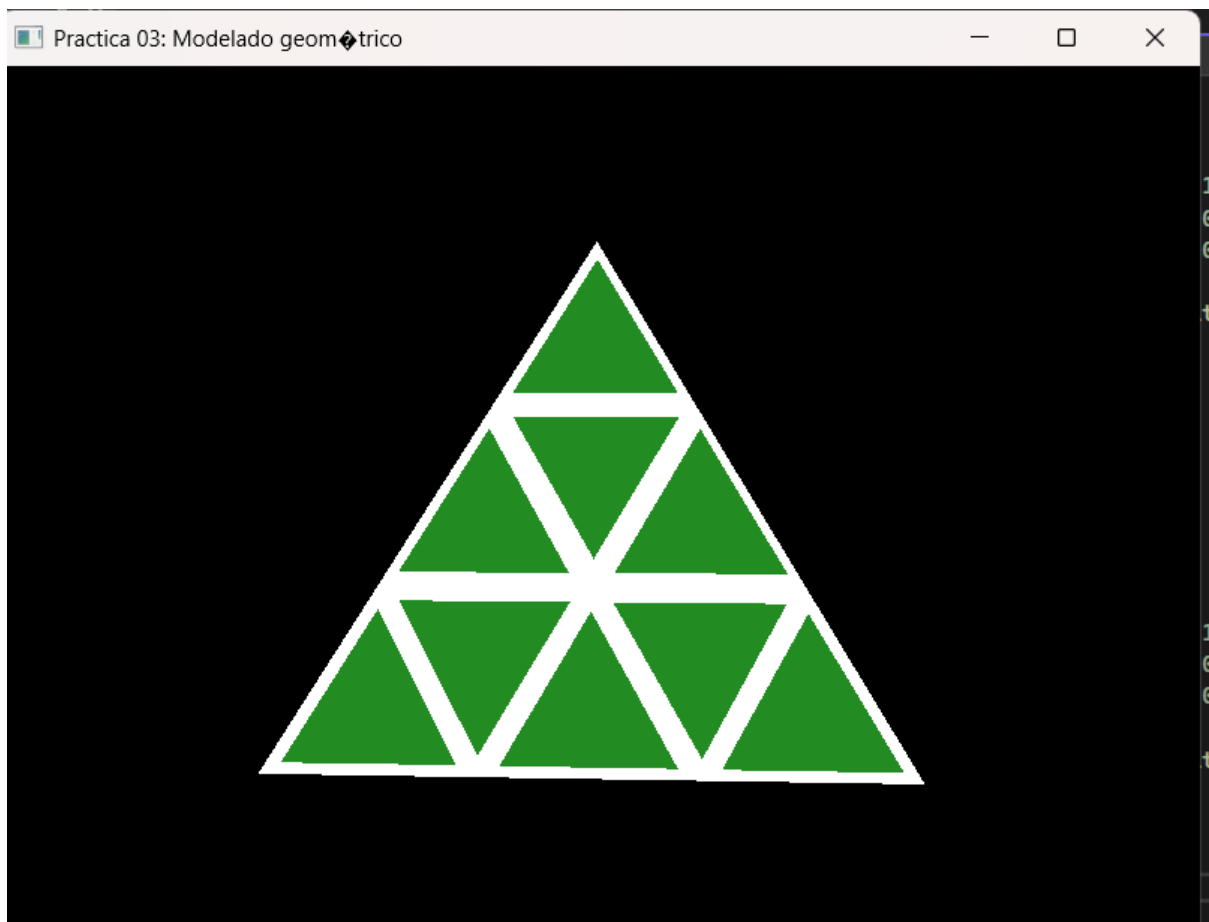
```

```

549
550     model = glm::mat4(1.0);
551     model = glm::rotate(model, -120 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
552     model = glm::translate(model, glm::vec3(0.43f, -0.13f, -0.42f));
553     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
554     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
555     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
556     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
557     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
558     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
559     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
560     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
561     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
562     meshList[0]->RenderMesh();
563

```

Y al ejecutar, esta cara se ve como a continuación.



El siguiente color que se modeló fue el color naranja con las siguientes coordenadas.


```

564 // Cara naranja
565 color = glm::vec3(0.282f, 0.820f, 0.800f);
566
567 model = glm::mat4(1.0);
568 model = glm::translate(model, glm::vec3(-0.28f, -0.01f, 0.43f));
569 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
570 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
571 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
572 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
573 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
574 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
575 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
576 meshList[0]->RenderMesh();
577
578 model = glm::mat4(1.0);
579 model = glm::translate(model, glm::vec3(-0.53f, -0.01f, 0.0f));
580 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
581 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
582 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
583 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
584 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
585 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
586 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
587 meshList[0]->RenderMesh();
588
589 model = glm::mat4(1.0);
590 model = glm::translate(model, glm::vec3(-0.03f, -0.01f, 0.86f));
591 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
592 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
593 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
594 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
595 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
596 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
597 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
598 meshList[0]->RenderMesh();
599

```

```

600 model = glm::mat4(1.0);
601 model = glm::translate(model, glm::vec3(-0.285f, 0.4f, 0.145f));
602 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
603 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
604 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
605 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
606 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
607 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
608 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
609 meshList[0]->RenderMesh();
610
611 model = glm::mat4(1.0);
612 model = glm::translate(model, glm::vec3(-0.03f, 0.4f, 0.575f));
613 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
614 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
615 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
616 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
617 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
618 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
619 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
620 meshList[0]->RenderMesh();
621
622 model = glm::mat4(1.0);
623 model = glm::translate(model, glm::vec3(-0.03f, 0.81f, 0.285f));
624 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
625 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
626 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
627 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
628 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
629 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
630 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
631 meshList[0]->RenderMesh();
632
633 model = glm::mat4(1.0);
634 model = glm::rotate(model, 120 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
635 model = glm::translate(model, glm::vec3(-0.175f, 0.29f, -0.28f));
636 model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
637 model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
638 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
639 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
640 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
641 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
642 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
643 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
644 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
645 meshList[0]->RenderMesh();

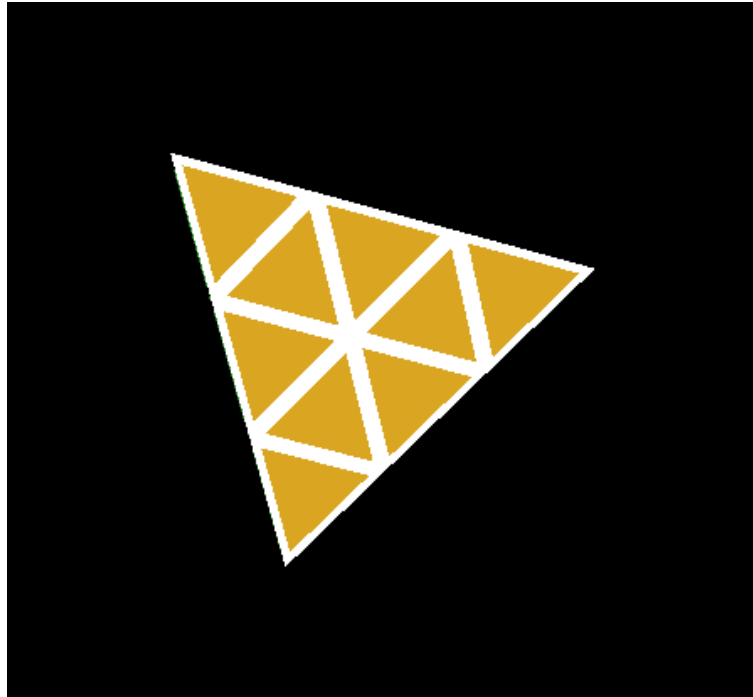
```

```

647     model = glm::mat4(1.0);
648     model = glm::rotate(model, 120 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
649     model = glm::translate(model, glm::vec3(0.07f, -0.13f, -0.42f));
650     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
651     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
652     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
653     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
654     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
655     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
656     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
657     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
658     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
659     meshList[0]->RenderMesh();
660
661     model = glm::mat4(1.0);
662     model = glm::rotate(model, 120 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
663     model = glm::translate(model, glm::vec3(-0.43f, -0.13f, -0.42f));
664     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
665     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
666     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
667     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
668     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
669     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
670     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
671     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
672     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
673     meshList[0]->RenderMesh();
674
675     model = glm::mat4(1.0);
676     model = glm::rotate(model, -120 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
677     model = glm::translate(model, glm::vec3(0.43f, -0.13f, -0.42f));
678     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 0.0f, 1.0f));
679     model = glm::rotate(model, -42 * toRadians, glm::vec3(1.0f, 0.0f, 0.0f));
680     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
681     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
682     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
683     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
684     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
685     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
686     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
687     meshList[0]->RenderMesh();

```

La cara resultante se muestra a continuación.



Otra de las caras que se hizo fue la cara color turquesa. Para esto se generaron las siguientes coordenadas.


```

689 // Cara turquesa
690 color = glm::vec3(0.855f, 0.647f, 0.125f);
691
692 model = glm::mat4(1.0);
693 model = glm::translate(model, glm::vec3(0.0f, -0.05f, -0.015f));
694 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
695 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
696 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
697 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
698 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
699 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
700 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
701 meshList[0]->RenderMesh();
702
703 model = glm::mat4(1.0);
704 model = glm::translate(model, glm::vec3(0.5f, -0.05f, -0.015f));
705 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
706 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
707 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
708 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
709 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
710 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
711 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
712 meshList[0]->RenderMesh();
713
714 model = glm::mat4(1.0);
715 model = glm::translate(model, glm::vec3(-0.5f, -0.05f, -0.015f));
716 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
717 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
718 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
719 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
720 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
721 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
722 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
723 meshList[0]->RenderMesh();
724

```

```

725 model = glm::mat4(1.0);
726 model = glm::translate(model, glm::vec3(-0.25f, -0.05f, 0.415f));
727 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
728 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
729 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
730 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
731 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
732 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
733 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
734 meshList[0]->RenderMesh();
735
736 model = glm::mat4(1.0);
737 model = glm::translate(model, glm::vec3(0.25f, -0.05f, 0.415f));
738 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
739 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
740 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
741 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
742 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
743 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
744 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
745 meshList[0]->RenderMesh();
746
747 model = glm::mat4(1.0);
748 model = glm::translate(model, glm::vec3(0.0f, -0.05f, 0.845f));
749 model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
750 model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
751 model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
752 model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
753 glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
754 glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
755 glUniform3fv(uniformColor, 1, glm::value_ptr(color));
756 meshList[0]->RenderMesh();
757

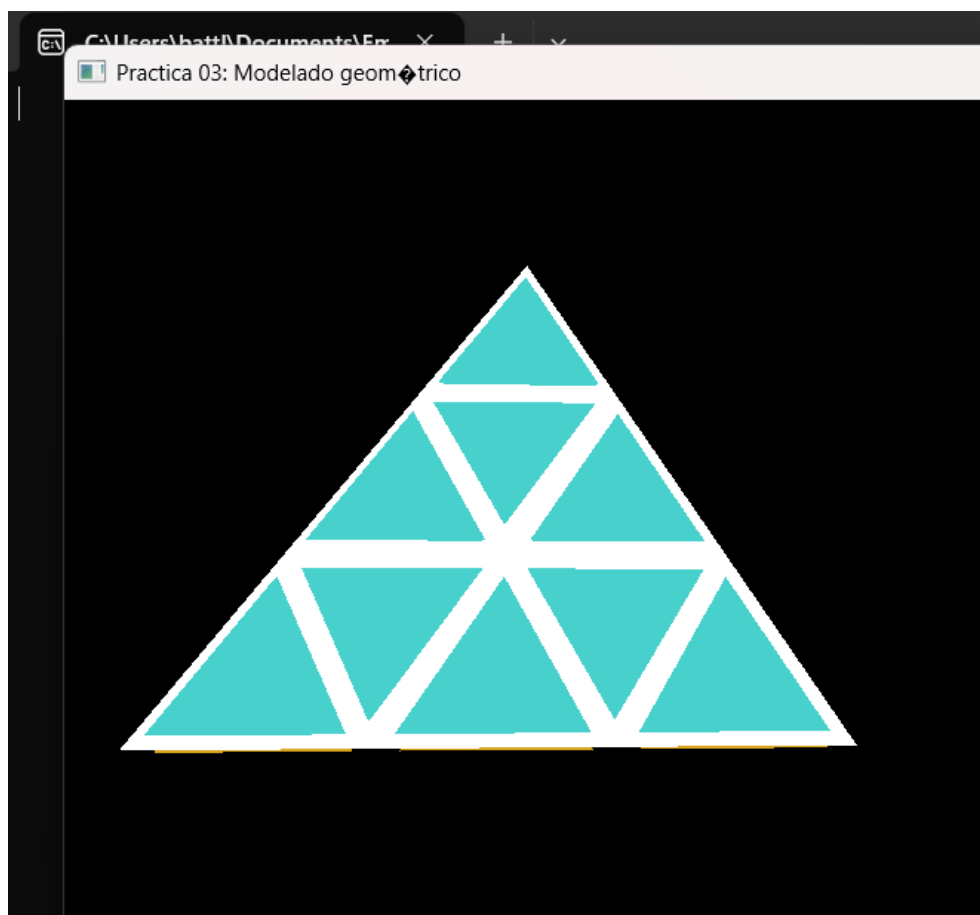
```

```

758     model = glm::mat4(1.0);
759     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
760     model = glm::translate(model, glm::vec3(0.25f, -0.05f, -0.015f));
761     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
762     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
763     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
764     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
765     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
766     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
767     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
768     meshList[0]->RenderMesh();
769
770     model = glm::mat4(1.0);
771     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
772     model = glm::translate(model, glm::vec3(-0.25f, -0.05f, -0.015f));
773     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
774     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
775     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
776     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
777     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
778     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
779     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
780     meshList[0]->RenderMesh();
781
782     model = glm::mat4(1.0);
783     model = glm::rotate(model, 180 * toRadians, glm::vec3(0.0f, 1.0f, 0.0f));
784     model = glm::translate(model, glm::vec3(0.0f, -0.05f, -0.445f));
785     model = glm::scale(model, glm::vec3(0.4f, 0.4f, 0.4f));
786     model = glm::rotate(model, glm::radians(mainWindow.getrotax()), glm::vec3(1.0f, 0.0f, 0.0f));
787     model = glm::rotate(model, glm::radians(mainWindow.getrotay()), glm::vec3(0.0f, 1.0f, 0.0f));
788     model = glm::rotate(model, glm::radians(mainWindow.getrotaz()), glm::vec3(0.0f, 0.0f, 1.0f));
789     glUniformMatrix4fv(uniformModel, 1, GL_FALSE, glm::value_ptr(model));
790     glUniformMatrix4fv(uniformView, 1, GL_FALSE, glm::value_ptr(camera.calculateViewMatrix()));
791     glUniform3fv(uniformColor, 1, glm::value_ptr(color));
792     meshList[0]->RenderMesh();

```

Y al momento de ejecutar obtenemos la siguiente vista.



2. Problemas

El mayor de los problemas presentados fue la ubicación en el espacio. Esto tiene más que ver con la parte de conocimientos previos de geometría analítica, así como también un enfoque más “prueba y error” que a el uso de matemáticas correctamente. Me atoró bastante el estar probando que valores podrían funcionar para lograr mostrar como se esperaba el ejercicio. Siento que existen formas más sencillas haciendo cálculos matemáticos, sin embargo, no cuento con la teoría o información necesaria para poder imaginar una forma más sencilla de hacerlo.

3. Conclusión

Siento que esta práctica ya tuvo un nivel mucho más alto a las anteriores y eso es retados al momento de aprender. Me gustó bastante y me tomó suficientes horas para completar. Es importante notar que creo que sería útil conocer algún atajo matemático para esto, ya que prácticamente son puras coordenadas, pero eso ya depende un poco más de mi y de mis antecedentes en asignaturas previas. Fuera de eso, creo que la explicación en clase y el ejercicio que se hizo ayudó bastante a entender lo que se debía entregar y a cómo llegar una solución posible.

4. Bibliografía

5. *Códigos de colores HTML*. (s. f.). HTML Color Codes.

<https://htmlcolorcodes.com/es/>